WATERLINE

Product requirements, design system and master build specification

Everything required to build the WATERLINE liquidity layer to production standard, with no gaps left to interpretation.

Companion to the product concept and execution plan. Covers domain model and architecture, the deterministic rule pack, every core flow, API behaviour, reliability engineering, edge cases, failsafe modes, backtesting and model governance, security and compliance, the complete design system, observability, testing, the production readiness checklist, and a master build prompt written to be executed without omission.

DOCUMENT	TRACK	STATUS	DATE
PRD v1.0	**Theme 2 — Creating Liquidity**	**Build-ready**	**15 October 2026**

PART 1

Product definition

WATERLINE is a liquidity layer embedded in du Business that converts approved invoices and idle balances into usable cash, priced deterministically and settled over licensed rails.

1.1 Problem statement

A UAE SME holding a buyer-approved invoice waits 45 to 90 days for cash it has already earned. The capital is not at risk and the amount is not in dispute; it is queuing. During the queue the business cannot act on opportunity or absorb shock. Conventional remedies — a bank facility or factoring — require the business to prove creditworthiness it has already demonstrated, on a timeline that misses the opportunity. The addressable inefficiency is the queue itself, not the SME's access to credit.

1.2 Users and jobs to be done

PERSONA	CONTEXT	JOB TO BE DONE
SME owner-operator	1–20 staff, no finance function, manages cash from a phone between operational tasks.	"Tell me what I can safely spend this week, and let me get paid now when I need to move."
SME finance lead	20–200 staff, part-time or single-person finance function, uses accounting software and spreadsheets.	"Give me a reliable forward view and let me close the gap without starting a loan application."
Anchor buyer AP clerk	Enterprise accounts payable, processes hundreds of invoices weekly, measured on cycle time and error rate.	"Let me confirm invoices in bulk and capture discounts without adding steps to my process."
du operations analyst	Monitors the funding book, resolves exceptions, answers SME queries.	"Show me what needs a human, why, and give me the authority to resolve it with a trail."
Risk and compliance officer	Accountable to CBUAE and internal audit for every automated decision.	"Prove to me that this decision was made by a rule I approved, and replay it on demand."
Funder credit officer	Deploys bank or fund capital against the receivables book.	"Show me loss performance by cohort, concentration, and the controls that bound my downside."

1.3 Scope

In scope, Phase 1	Dynamic discounting on buyer-approved receivables; milestone escrow via programmable mandate at a licensed bank; treasury autopilot into low-risk short-term instruments; cash-flow forecasting; buyer confirmation portal; dispute workflow; operator console; SME mobile and web experience embedded in du Business; Arabic and RTL.
In scope, Phase 2	Open Finance account information and payment initiation at depth; e-invoice verification at source; multi-funder allocation engine; anchor-buyer ERP connectors; supplier-initiated early payment requests.
Phase 3, flagged	Tokenised recurring revenue for upfront capital, pending CBUAE and VARA guidance. Fully specified, feature-flagged, zero dependency from Phases 1 and 2.

Explicitly out of scope	Unsecured lending of any kind; consumer credit; FX and cross-border payments; crypto custody or trading; insurance; any write path into du BSS; any participation in rating or charging.

1.4 Product principles

- **No model decides who gets paid.** Machine learning forecasts and ranks; a deterministic versioned rule pack decides and prices. Always.
- **Every decision is replayable.** Input hash, rule-pack version, resolved parameters, outcome, plain-language reason — written append-only at decision time, not reconstructed.
- **Show the arithmetic.** The SME sees days early, rate, fee in dirhams and net proceeds before accepting, and the same numbers afterwards.
- **Degrade, never fail.** Five declared modes. Partial service with honest status beats a blank screen.
- **The SME's floor is absolute.** Treasury autopilot never breaches the minimum balance the SME sets, under any optimisation.
- **Exceptions surface, never silently fail.** An SME must never learn about a problem by noticing money has not arrived.
- **One binary.** Pilot and full base run identical code behind flags. No pilot build to throw away.

PART 2

Architecture

2.1 Service topology

Twelve services, each independently deployable, communicating over authenticated internal APIs for synchronous work and an event log for asynchronous work. All services on the funding path are stateless.

SERVICE	RESPONSIBILITY	PATH
identity	SME and staff identity, UAE Pass and du SSO federation, session, entitlement, role.	Hot
consent	Scoped, revocable consent capture; purpose registry; enforcement hooks; revocation propagation.	Hot
registry	Entities, trade licences, UBO, bank accounts, buyer relationships, KYB state.	Warm
invoice	Invoice ingestion, normalisation to the Peppol model, hashing, duplicate detection, verification state machine.	Warm
ruleengine	Deterministic evaluation of the active rule pack. Pure function: no I/O during evaluation, no clock access, no randomness.	Hot
forecast	Cash-flow forecasting, obligation detection, risk ranking, anomaly scoring. Advisory only; produces features, never decisions.	Warm
decision	Orchestrates a decision: assembles features, calls ruleengine, writes the immutable decision record, emits the event.	Hot
ledger	Double-entry ledger of record. Every movement is a balanced transaction; balances are derived, never stored as truth.	Hot
payout	Money movement orchestration over du Pay and Aani. Idempotent, signed, reconciled.	Hot
escrow	Milestone contract lifecycle, condition evaluation, release instruction to the custodian bank.	Warm
treasury	Sweep and recall engine, floor enforcement, obligation calendar, instrument allocation.	Warm
audit	Append-only decision and event store, retention, legal hold, regulatory retrieval.	Cold

2.2 Hot, warm and cold paths

Separating paths by latency budget is what keeps analytics load from ever contending with a payout.

PATH	BUDGET AND STORAGE	CARRIES
Hot	p99 under 400 ms. In-memory cache plus primary Postgres. Served active-active from both zones.	Authentication, balance display, limit and price quote, offer acceptance, payout initiation, status.
Warm	Seconds to minutes. Queue-driven, primary Postgres plus search index.	Invoice verification, buyer confirmation, forecast refresh, escrow condition evaluation, sweep calculation, reconciliation, notification.
Cold	Minutes to hours. Object storage, columnar warehouse, immutable audit store.	Backtesting, shadow pricing, portfolio analytics, regulatory retrieval, statement generation, model training.

A quote is served hot from precomputed eligibility and a cached rule pack. Nothing on the hot path calls an external provider synchronously: if fresh bank data is needed, the quote is served from last-known-good with an explicit freshness indicator, and the refresh happens warm.

2.3 Multi-tenancy and partitioning

- **Tenant key on every row.** Enforced by Postgres row-level security, not by application discipline. A query without tenant context returns nothing.
- **Hash partitioning** of high-volume tables (invoice, decision, ledger_entry, event) by tenant, so account growth does not degrade per-account performance.
- **Per-tenant quotas** on request rate, concurrent decisions and queue depth, so no single tenant can consume shared capacity.
- **Connection pooling per partition group** with bulkheads, so saturation in one group cannot exhaust the pool for others.

2.4 Active-active

Two UAE availability zones serve live traffic simultaneously. Both are primary for reads and writes on the hot path.

Data	Synchronous replication for the ledger and decision store — correctness over latency, no exceptions. Asynchronous replication for search, cache and analytics.
Conflict policy	Money movement is partitioned by tenant to a home zone, so two zones never write the same ledger account concurrently. On failover the home zone is reassigned, never shared.
Objectives	RPO zero for ledger and decision records. RTO 15 minutes for full service, 2 minutes for degraded read-only.
Drill cadence	Zone failover drilled monthly in staging and quarterly in production during a declared window, with the result published internally. An undrilled failover is assumed broken.

2.5 Core data model

```
entity          id, tenant_id, legal_name, trade_licence, licence_expiry,
                emirate, sector_code, kyb_state, ubo[], created_at
buyer_link      id, tenant_id, entity_id, buyer_entity_id, relationship_age_days,
                confirmed_invoice_count, avg_days_to_pay, dispute_count, tier
invoice         id, tenant_id, entity_id, buyer_link_id, external_ref,
                issue_date, due_date, currency, gross_amount, tax_amount,
                invoice_hash, source(einvoice|buyer_api|upload|du_ap),
                verification_state, verified_at, peppol_doc_ref
decision        id, tenant_id, subject_ref, decision_type, input_hash,
                rulepack_version, feature_snapshot_ref, outcome,
                limit_amount, price_bps_per_day, reason_code, reason_text,
                evaluated_at, actor, shadow(bool)
offer           id, tenant_id, invoice_id, decision_id, advance_rate,
                fee_amount, net_proceeds, settlement_date, expires_at, state
ledger_entry    id, tenant_id, txn_id, account_code, direction, amount,
                currency, value_date, posted_at, source_event_id
payout          id, tenant_id, offer_id, idempotency_key, rail(dupay|aani),
                destination_ref, amount, state, provider_ref, attempts,
                last_error_code, reconciled_at
escrow_contract id, tenant_id, buyer_entity_id, supplier_entity_id,
                custodian_ref, total_amount, milestones[], state
milestone       id, contract_id, sequence, description, amount,
                condition_type, evidence_ref, approved_by, approved_at, state
sweep           id, tenant_id, direction(out|in), amount, instrument_ref,
                floor_at_decision, obligation_window_json, state
consent_grant   id, tenant_id, entity_id, purpose, scope[], granted_at,
                expires_at, revoked_at, provider, provider_consent_ref
event           id, tenant_id, type, payload_ref, occurred_at, sequence,
                caused_by_event_id, idempotency_key
```

Invariants enforced in the database, not the application: every `txn_id` in `ledger_entry` sums to zero across directions; `invoice_hash` is unique per tenant and globally indexed for cross-tenant duplicate detection; `decision` rows are insert-only with no update or delete grant on any application role; `payout.idempotency_key` is unique per tenant.

PART 3

The rule pack

The rule pack is the single place where eligibility, limits and prices are decided. It is data, not code, and it is versioned, signed, replayable and promotable independently of any deployment.

3.1 Structure

```
rulepack:
  version: "2026.10.3"              # immutable once promoted
  signed_by: "risk-committee"
  effective_from: "2026-10-15T00:00:00Z"
  eligibility:
    - id: E01  when: entity.kyb_state != 'verified'        then: reject(E_KYB)
    - id: E02  when: entity.licence_expiry < today + 30d    then: reject(E_LICENCE)
    - id: E03  when: invoice.verification_state != 'verified' then: reject(E_UNVERIFIED)
    - id: E04  when: invoice.due_date < today               then: reject(E_OVERDUE)
    - id: E05  when: invoice.gross_amount < 2000 AED       then: reject(E_MIN_SIZE)
    - id: E06  when: buyer_link.tier == 'unrated'
                 and buyer_link.confirmed_invoice_count < 3 then: cap(advance_rate, 0.60)
  limits:
    base:        min(0.90 * invoice.gross, entity_cap, buyer_cap, sector_cap)
    entity_cap:  f(tenure_months, billing_behaviour_band, confirmed_history)
    buyer_cap:   f(buyer_tier, avg_days_to_pay, dispute_rate)
    sector_cap:  table(sector_code)
    portfolio:   hard_stop if book_exposure > portfolio_ceiling
  pricing:
    rate_bps_per_day: base_curve(buyer_tier) + risk_adj(features) + funding_cost
    floor_bps: 1.2   ceiling_bps: 9.0
    fee = net_days_early * rate_bps_per_day * principal / 10000
  explanations:
    E_KYB:        "We need your trade licence verified before we can price this."
    E_LICENCE:    "Your trade licence expires within 30 days. Renew to continue."
    E_UNVERIFIED: "Your buyer hasn't confirmed this invoice yet."
    ...
```

3.2 Determinism guarantees

- **Pure evaluation.** `ruleengine` performs no I/O, reads no clock and uses no randomness. Time and all features are passed in as an explicit, hashed feature snapshot.
- **Hashed inputs.** The feature snapshot is canonicalised and hashed; the hash is stored on the decision. Replaying the same hash against the same pack version must produce a byte-identical outcome, and this is asserted in CI.
- **No silent fallback.** A missing feature is a typed rejection with a reason code, never a default value substituted quietly.
- **Decimal arithmetic.** All money is integer minor units; all rates are fixed-point. No floating point touches a monetary calculation anywhere in the system.

3.3 Promotion lifecycle

1 — Author	Change proposed as a diff against the active pack, with rationale and expected population impact.
2 — Shadow	Candidate evaluated against 12 months of history and 30 days of live traffic in parallel. Outcome deltas reported by segment: approvals gained and lost, price moves, limit moves, expected loss change.

3 — Review	Risk committee signs the pack. Fairness dispersion across sector, size and emirate is reviewed as a required gate.
4 — Canary	Promoted to 5% of tenants for 72 hours with automatic rollback on guardrail breach.
5 — Promote	Full promotion. The prior pack remains loadable forever for replay; nothing is ever deleted.
Rollback	Pointer change to the previous version, effective within 60 seconds, requiring no deployment.

3.4 Decision record


```
{ "decision_id":"dec_01J...", "tenant_id":"t_8821", "subject_ref":"inv_7741",
  "decision_type":"advance_offer", "rulepack_version":"2026.10.3",
  "input_hash":"sha256:9f2c...", "feature_snapshot_ref":"s3://audit/fs/9f2c...",
  "outcome":"approved", "limit_amount":4200000, "price_bps_per_day":3.4,
  "binding_rule":"L02:buyer_cap", "reason_code":"OK_CAPPED_BY_BUYER",
  "reason_text":"Approved at AED 42,000. Capped by this buyer's limit, which
                 grows as they settle more invoices on time.",
  "evaluated_at":"2026-10-15T09:14:22.881Z", "actor":"system", "shadow":false }
```


Written append-only before the offer is shown to the SME. If the write fails, the offer is not shown — the audit record is a precondition of the decision, not a side effect of it.

PART 4

Core flows

4.1 Onboarding and consent

Target: under three minutes, zero documents uploaded in the standard path.

01 **Entry**

SME taps WATERLINE in du Business. Already authenticated via du SSO; no new credential is created.

02 **Identity**

UAE Pass step-up for the principal. Trade licence and establishment card verified against registry data. KYB state moves to `verified` or into a typed exception queue.

03 **Consent**

Scoped consent presented in plain language, one purpose per toggle: billing history, bank data via Open Finance, buyer confirmation contact. Each is independently revocable and each shows exactly what it unlocks.

04 **First value**

Before any invoice is uploaded, the SME sees its indicative limit derived from du billing history alone. This is the activation moment and it requires nothing from the user.

Revocation propagates within 60 seconds: dependent features disable immediately, derived features are purged within 24 hours, and existing funded positions continue to their contractual conclusion with that fact disclosed at revocation time.

4.2 Invoice verification state machine

```
received ──normalise──> parsed ──hash──> deduped
deduped ──einvoice match──>      verified
deduped ──buyer confirm──>       verified
deduped ──du AP file match──>    verified
deduped ──no route──>            unverifiable(reason)
parsed  ──hash collision──>      duplicate(ref to original)
verified ──buyer disputes──>     disputed ──resolve──> verified | written_off
verified ──buyer amends──>       superseded(new invoice id)
```

Hashing	Canonical hash over buyer tax ID, supplier tax ID, invoice number, gross amount and issue date. Resistant to reformatting and whitespace, sensitive to any material change.
Cross-tenant check	The hash index is global. The same receivable presented by two tenants is caught even though the data is tenant-isolated, with the collision routed to fraud review and never auto-rejected.
Verification priority	E-invoice registry, then du AP file, then buyer API callback, then buyer portal single-click. Weaker routes carry a lower advance-rate cap set in the rule pack.

4.3 Pricing, acceptance and payout

01 **Quote**

`decision` assembles the feature snapshot, calls `ruleengine`, writes the decision record, returns limit and rate. Budget: p99 under 400 ms, served without any synchronous external call.

02 **Model**

The SME moves a date slider and sees fee and net proceeds recompute locally against the quoted rate curve. No server round trip, so the interaction is instant and works on poor connectivity.

03 **Accept**

Offer accepted with the decision ID and a client idempotency key. The offer is re-validated server-side against its expiry and the current pack; a changed price is surfaced explicitly and requires fresh acceptance, never applied silently.

04 **Reserve and post**

Funding source reserved, assignment recorded, ledger transaction posted. Ledger commit and payout initiation are bound by an outbox so that a committed ledger entry always has exactly one payout attempt.

05 **Settle**

Payout dispatched to du Pay or Aani with the idempotency key. Terminal success, terminal failure or timeout; timeouts are resolved by reconciliation rather than by retrying blindly.

06 **Notify and reconcile**

SME notified through du's gateway. Daily statement reconciliation to the dirham, with any break escalated same day.

4.4 Escrow lifecycle

Funds are held at a licensed custodian bank under a programmable mandate. WATERLINE never holds client money.

Award	Buyer funds the contract at the custodian. Supplier sees confirmed, locked funds before starting work — the single most valuable moment in the product.
Milestone conditions	Three types: explicit buyer approval, document-evidenced delivery with buyer review window, and time-based release with buyer veto. Each is a declared condition in the contract, evaluated by `escrow` and instructed to the custodian.
Release	On condition satisfaction, release is instructed automatically. Buyer inaction inside a declared review window is itself a declared condition, which removes the most common cause of stalled milestone payments.
Dispute	Either party raises a dispute; the disputed milestone freezes while undisputed milestones continue. Resolution is logged with evidence and the state transition is auditable.
Termination	Unreleased funds return to the buyer under the contract's declared termination terms, with the full milestone history exportable for any downstream proceeding.

4.5 Treasury autopilot

The SME sets one number: the minimum balance it will never drop below. Everything else is derived.

Obligation calendar	Built from payroll cadence, rent, VAT deadlines, recurring supplier payments and du's own billing cycle, plus any obligation the SME adds manually. Forward-looking, not inferred from the past alone.
Sweepable amount	`balance − floor − obligations(horizon) − safety_margin`. Safety margin scales with forecast error for that tenant, so a volatile business is swept more conservatively.
Instruments	Low-risk short-term only, with a published maximum maturity and same-day or next-day liquidity. Instrument eligibility is rule-pack data, so it is reviewable and changeable without a deployment.
Recall	Automatic ahead of every detected obligation with a margin. A recall that would arrive late is never initiated; the sweep that would have required it is not made in the first place.
Hard guarantee	No optimisation, no yield opportunity and no operator override may breach the floor. This is enforced in `ruleengine` as a hard stop, and it is the single most heavily tested invariant in the system.

PART 5

API behaviour

5.1 Conventions

Transport	HTTPS only, TLS 1.3 minimum. Internal service-to-service uses mutual TLS with short-lived certificates issued by an internal CA.
Auth	OAuth 2.1 client credentials for partner and service clients; OIDC with UAE Pass step-up for SME principals. Tokens are audience-scoped, maximum 15-minute lifetime, never passed in query strings.
Versioning	Path major version (`/v1/`) plus a date-pinned minor via `X-API-Version` . Breaking changes require a new major; two majors are supported concurrently for a minimum of 12 months.
Idempotency	`Idempotency-Key` is **required** on every POST, PUT and PATCH that moves money or creates a binding obligation. Keys are client-generated UUIDv7, scoped per tenant, retained 24 hours. Replay returns the original response with `Idempotent-Replay: true` .
Pagination	Opaque cursor only; offset pagination is not offered. Default page 50, maximum 200. Cursors encode a stable sort key and are valid 24 hours.
Rate limits	Token bucket per tenant per endpoint class. Responses always carry `RateLimit-Limit` , `RateLimit-Remaining` , `RateLimit-Reset` . 429 always carries `Retry-After` .
Errors	RFC 9457 `application/problem+json` with a stable machine `code` , human `detail` , `trace_id` and, where applicable, `retryable: true|false` . Error codes are a public contract and never repurposed.
Time and money	All timestamps RFC 3339 UTC with milliseconds. All amounts integer minor units plus explicit ISO-4217 currency. Never a float, never a bare number.
Tracing	`traceparent` accepted and propagated end to end. Every response returns `trace_id` , including errors, so support can locate any transaction from a screenshot.

5.2 Endpoint catalogue

METHOD AND PATH	PURPOSE	IDEM.	P99 BUDGET
`POST /v1/entities`	Register an SME entity and begin KYB	Required	800 ms
`POST /v1/consents`	Grant scoped consent for a declared purpose	Required	400 ms
`DELETE /v1/consents/{id}`	Revoke consent, propagating within 60 s	n/a	400 ms
`POST /v1/invoices`	Submit an invoice for verification	Required	600 ms
`POST /v1/invoices/{id}/confirm`	Buyer confirms amount and due date	Required	600 ms
`GET /v1/invoices/{id}/quote`	Deterministic limit and price quote	n/a	400 ms
`POST /v1/offers/{id}/accept`	Accept an offer and trigger funding	Required	1200 ms
`GET /v1/payouts/{id}`	Payout state and provider reference	n/a	300 ms
`POST /v1/escrow/contracts`	Create a milestone escrow contract	Required	900 ms
`POST /v1/escrow/milestones/{id}/approve`	Approve a milestone, triggering release	Required	900 ms

METHOD AND PATH	PURPOSE	IDEM.	P99 BUDGET
`PUT /v1/treasury/policy`	Set floor, horizon and instrument preferences	Required	500 ms
`GET /v1/treasury/projection`	Forward cash projection with obligations	n/a	700 ms
`GET /v1/decisions/{id}`	Full decision record with reason and pack version	n/a	400 ms
`POST /v1/decisions/{id}/replay`	Re-evaluate a stored input hash for audit	Required	2000 ms
`GET /v1/statements/{period}`	Signed period statement, cold path	n/a	5000 ms

5.3 Error taxonomy

HTTP	CODE	MEANING	CLIENT ACTION
400	`invalid_request`	Schema or semantic validation failed	Fix and resubmit. Never retry unchanged.
401	`unauthenticated`	Token missing, expired or wrong audience	Refresh token, retry once.
403	`consent_required`	Operation needs a consent that is absent or revoked	Drive consent flow. Do not retry.
404	`not_found`	Resource absent or outside tenant scope	Do not retry.
409	`idempotency_conflict`	Key reused with a different payload	Bug in the client. Alert, do not retry.
409	`offer_expired`	Offer lapsed or pack changed since quote	Re-quote and present the new price explicitly.
409	`duplicate_invoice`	Invoice hash already present	Surface the original. Route to review if cross-tenant.
422	`rule_rejected`	Deterministic rejection with reason code	Show the plain-language reason. Never retry.
423	`kill_switch_active`	Funding suspended by an operator or guardrail	Show degraded status. Poll status endpoint.
429	`rate_limited`	Quota exhausted	Honour `Retry-After` exactly.
500	`internal_error`	Unexpected failure, already alerted	Retry with backoff. Safe only with idempotency key.
502/504	`upstream_unavailable`	Provider down or timed out	Retry with backoff; reconcile if money may have moved.
503	`degraded`	Service in a declared degrade mode	Read `degrade_mode` and adapt the UI.

5.4 Webhooks

- **Signed** with HMAC-SHA256 over timestamp and raw body, in `WATERLINE-Signature`, with a 5-minute tolerance window to bound replay.
- **At-least-once** delivery. Consumers must dedupe on `event_id`, and this requirement is stated in the contract rather than assumed.
- **Ordered per subject** via a monotonic `sequence`. Consumers must tolerate reordering across subjects and are given the sequence to do so.
- **Retries** for 24 hours on exponential backoff, then to a dead-letter queue with operator notification and a manual replay endpoint.
- **No sensitive payloads.** Events carry references, not financial detail; the consumer fetches detail over the authenticated API.

PART 6

Reliability engineering

6.1 Retries and exponential backoff

One retry policy, defined centrally, applied by every client. Per-call overrides are a review finding, not a convenience.

```
delay(n) = min(cap, base * 2^n) * jitter      jitter ~ U(0.5, 1.5)   # full jitter

class          base    cap      max_attempts  budget     retry_on
hot_read       50ms    800ms    3             1.5s       503,504,timeout
quote          80ms    1s       2             2s         503,504,timeout
payout_init    250ms   8s       5             45s        502,503,504,timeout
provider_poll  1s      60s      12            15m        any non-terminal
webhook_out    5s      1h       14            24h        non-2xx
batch_job      30s     15m      8             6h         transient only
```

Full jitter, always	Equal-jitter and no-jitter schedules synchronise retries across clients and produce a thundering herd precisely when a dependency is already struggling.
Retry budget	A client may spend at most 10% of its request volume on retries in any 60-second window. Exceeding the budget opens the circuit rather than amplifying load into a failing dependency.
Never retry non-idempotent	A money movement without an idempotency key is not retried under any circumstance. It is reconciled instead.
Deadline propagation	The remaining deadline travels in the request context. A retry that cannot complete within the deadline is not attempted — it would burn the dependency for a result nobody will read.

6.2 Idempotency and deduplication

Request level	`Idempotency-Key` stored with a fingerprint of the canonicalised request body. Same key and same fingerprint replays the stored response; same key and different fingerprint returns 409. In-flight duplicates block on the original rather than racing it.
Ledger level	`(tenant_id, source_event_id)` is unique on `ledger_entry`. A double-posted event is rejected by the database, not by hopeful application logic.
Payout level	The provider-facing idempotency key is derived deterministically from `offer_id`, so a retry after an ambiguous timeout cannot produce a second payment even across a process restart.
Event consumption	Consumers write `(consumer, event_id)` to a processed-events table inside the same transaction as their side effect. Exactly-once effect from at-least-once delivery.
Invoice level	Global hash index catches the same receivable across tenants and across time, including after reformatting.
Notification level	Content hash plus a 10-minute window suppresses duplicate SMS and push, so a retry storm never becomes a message storm for the SME.

6.3 Queues and asynchronous work

```
topic                    partition key    ordering        DLQ   consumers
invoice.submitted        tenant_id        per tenant      yes   verify, fraud
invoice.verified         invoice_id       per invoice     yes   pricing, notify
decision.recorded        tenant_id        per tenant      yes   audit, analytics
offer.accepted           offer_id         strict/offer    yes   funding
ledger.posted            txn_id           strict/txn      yes   payout, recon
payout.state_changed     payout_id        per payout      yes   notify, recon
escrow.condition_met     contract_id      per contract    yes   release
treasury.obligation_due  tenant_id        per tenant      yes   recall
webhook.outbound         subscription_id  per subscriber  yes   dispatcher
```

- **Transactional outbox everywhere.** Events are written to an outbox table in the same transaction as the state change, then published by a relay. A state change without its event, or an event without its state change, is structurally impossible.
- **Dead-letter discipline.** Every topic has a DLQ. A DLQ that is non-empty for more than 15 minutes pages a human. DLQ messages are replayable individually or in bulk after a fix, with replay itself idempotent.
- **Poison-message handling.** Three consecutive failures on the same message routes it to the DLQ rather than blocking the partition. Partition-blocking is the most common cause of silent queue stalls and is designed out.
- **Backpressure.** Queue depth above threshold sheds load at the edge for low-priority work (analytics, non-critical notification) before any funding-path work is affected. Priority is declared per topic, not decided during an incident.
- **Ordering honesty.** Strict ordering only where it is genuinely required — per offer, per transaction, per contract. Everything else is explicitly unordered so consumers are written to tolerate it.

6.4 Caching

LAYER	CONTENTS	TTL	INVALIDATION
Edge CDN	Static assets, fonts, icon sprite	1 year, content-hashed	Immutable; new hash on change
Client	Rule-pack rate curve for local fee modelling	Offer lifetime	Discarded on offer expiry
API response	Projections, statements, portfolio reads	30–300 s	ETag plus event-driven purge
Feature cache	Assembled feature snapshots for quoting	120 s	Purged on any source event
Rule pack	Parsed, compiled active pack	Until version change	Pointer change broadcast to all instances
Reference data	Sector tables, instrument eligibility, holiday calendar	1 hour	Scheduled plus manual purge
Negative cache	Known-absent lookups	10 s	Short by design, to bound staleness

- **Never cached:** balances, limits at the moment of acceptance, consent state, KYB state, kill-switch state. These are read through to the primary every time, without exception.
- **Stampede protection:** single-flight per key plus probabilistic early expiry, so a popular key never produces a synchronised refresh across instances.
- **Stale-while-revalidate** for read-only analytics only, never for anything an SME acts on financially.
- **Cache-key hygiene:** tenant ID is part of every key. A key that could be shared across tenants fails the build.

6.5 Error handling discipline

"Try/catch" is a policy here, not a reflex. The rules are explicit because inconsistent error handling is how financial systems lose money quietly.

Typed errors only	Every catch block handles a declared error type. A bare catch-all is permitted exactly once per service, at the top-level request boundary, where it logs with the trace ID, increments a counter, and returns `internal_error`.
Never swallow	A caught error is either handled with a declared recovery, or rethrown with added context. Logging and continuing is prohibited on any path that touches money.
Fail closed on money	Ambiguity on the funding path resolves to not paying, then reconciling. Ambiguity on a read path resolves to showing last-known-good with a freshness indicator.
Compensation, not rollback	Distributed operations use explicit compensating transactions — reverse the ledger entry, void the reservation, release the escrow hold — written and tested as first-class code paths, not as cleanup scripts.
Error context	Every error carries tenant, subject reference, decision ID where applicable, and trace ID. An error without context is a defect raised in code review.
Timeouts everywhere	No network call, database query or lock acquisition is made without an explicit timeout. A missing timeout fails the build via lint rule.

6.6 Circuit breakers and bulkheads

```
breaker(dependency):
  closed   → open     when error_rate > 50% over 20 requests in 10s
                       or p99 latency > 3x baseline for 30s
  open     → half     after 30s cooldown
  half     → closed   after 5 consecutive successes
  half     → open     on any failure

bulkheads: dupay=40 conns  aani=40  openfinance=20  einvoice=15
           custodian=10    notification=25  (per zone, independent pools)
```

Each dependency has an isolated connection pool and concurrency limit, so an Open Finance provider degrading cannot exhaust the pool that payouts need. Breaker state is exposed on the status endpoint and drives the UI's degrade messaging directly, so what the SME sees is derived from reality rather than guessed.

PART 7

Edge cases

The cases below are specified because they are the ones that produce either a loss or a furious SME. Each has a defined behaviour, not a best effort.

INVOICE AND VERIFICATION

Buyer amends after funding	Original invoice is superseded, not edited. If the amount falls below the advance, the shortfall is recovered under the assignment terms disclosed at acceptance; the SME is notified before any recovery, never after.
Credit note issued post-funding	Netted against the receivable. If it exceeds the outstanding balance, the excess becomes a recoverable with a payment plan rather than an immediate demand.
Partial payment by buyer	Applied to the funded portion first. The remainder stays open; the invoice does not close until fully settled, and the SME sees both figures.
Buyer pays the SME directly	Detected by Open Finance inflow matching. SME is notified with a clear repayment instruction and a grace window before any fee. Repeat occurrences reduce the advance rate via the rule pack.
Duplicate across tenants	Both submissions freeze. Neither is auto-rejected. Routed to fraud review with both tenants notified that verification is in progress — never accused.
Invoice currency not AED	Rejected in Phase 1 with an explicit reason. FX is out of scope and the product says so rather than silently mishandling it.
Due date in the past at submission	Rejected as overdue with a reason, and routed to collection support rather than financing.
Buyer entity dissolved	Registry check at verification and re-check at settlement. Funding blocked; existing exposure moves to recovery with the SME informed immediately.

PAYMENT AND SETTLEMENT

Payout timeout, outcome unknown	Never retried blind. Status is polled using the deterministic idempotency key until terminal. The SME sees "sending" with an honest expected time, not a false success.
Payout succeeds, callback lost	Reconciliation detects the orphan within the cycle and completes the state transition. The ledger is authoritative; the provider callback is a convenience.
Destination account closed	Payout fails cleanly, ledger entry is compensated, SME is prompted to update destination, and the offer is re-honoured at the original price for 48 hours.
Aani unavailable	Fallback to du Pay internal balance where destination permits. Otherwise the offer is held at the quoted price and auto-retried, with the SME told exactly what is happening.
Duplicate payout detected	Impossible by key construction. If one is ever observed, the funding kill switch engages automatically and the event pages immediately — treated as a sev-1 integrity breach.
Weekend or holiday	Aani settles 24/7, so there is no batch window. Obligation calendars still respect UAE public holidays for recall timing.
Rounding	Integer minor units throughout. Any residual fil is posted to a declared rounding account, reconciled daily, and never absorbed silently.

LIMITS, PRICING AND CONSENT

Pack changes between quote and accept	Acceptance re-validates. A changed price is shown explicitly and requires fresh acceptance. A price never changes silently between the screen and the money.
Limit drops mid-session	Existing accepted offers are honoured. New quotes reflect the new limit with the reason stated — "two invoices moved past 30 days overdue" — rather than an unexplained reduction.

Concurrent acceptance exceeding limit	Serialised per tenant with a reservation held during funding. The second acceptance is rejected with `limit_exceeded` and the available headroom shown.
Consent revoked mid-transaction	In-flight funding completes; no new decision is made. Dependent features disable within 60 seconds and derived features purge within 24 hours.
Trade licence expires while funded	New funding blocked at 30 days before expiry with escalating notice. Existing positions run to conclusion. No retroactive action is taken against a compliant business.
Feature unavailable at quote time	Typed rejection with a reason. No default is substituted. A quote is never issued on incomplete inputs.

TREASURY

Unforeseen large outflow after sweep	Emergency recall on the fastest instrument path, with the SME shown the exact arrival time. If the arrival would be late, overdraft is never created — the shortfall is surfaced with options instead.
Payroll date moves	Calendar updates and recall re-plans immediately. Payroll detection uses a conservative window precisely because the date moves in practice.
Instrument becomes illiquid	Instrument is frozen for new sweeps, existing holdings are run to maturity, and the SME is notified with the maturity date. Eligibility is rule-pack data, so removal needs no deployment.
Floor set above balance	Valid state. Nothing is swept, no error is shown, and the SME is simply told no sweep is possible at the current floor.
Forecast degrades for a tenant	Safety margin widens automatically with forecast error, so a volatile business is swept more conservatively without any manual intervention.

ESCROW

Buyer never approves a met milestone	The declared review window expires and release proceeds under the time-based condition with buyer veto. This is agreed at contract creation, which is what makes it enforceable and fair.
Contract abandoned mid-way	Unreleased funds return under declared termination terms. Full milestone history is exportable for any downstream proceeding.
Partial milestone delivery	Supported only where the contract declares divisible milestones at creation. Otherwise the milestone is binary, which prevents disputes about what "mostly done" means.
Custodian unavailable at release	Release is queued with the condition satisfaction timestamped at the moment it was met, not at the moment the instruction succeeded. The supplier is not penalised for infrastructure.

PART 8

Failsafe modes

Five declared modes. Transitions are automatic on guardrail breach and manual by operator command. The current mode is published on the status endpoint and drives the interface directly.

MODE	TRIGGER	BEHAVIOUR	SME SEES
M0 Normal	—	All engines active, straight-through processing at target.	Full product.
M1 Cautious	Loss rate, dispute rate or forecast error crosses a warning band; or a dependency breaker is open.	Advance rates reduced by a rule-pack delta, new buyer relationships capped harder, manual review threshold lowered.	Normal product with slightly lower limits and an honest note explaining the temporary tightening.
M2 Quote-only	Payout rail unavailable, or reconciliation break unresolved past threshold.	Quotes continue and are honoured at the quoted price. Acceptances queue. No money moves.	"We can price this now and pay as soon as the rail is back" with a live status and the price locked.
M3 Funding suspended	Integrity breach, duplicate payout observed, portfolio ceiling breach, or regulatory instruction.	No new funding of any kind. Existing positions, collections, escrow releases and treasury recalls continue normally.	Clear suspension notice with expected review time. Existing money keeps working.
M4 Read-only	Primary datastore degraded, or zone failover in progress.	Reads from replica with a freshness indicator. All writes rejected with `degraded`.	Balances and history with a visible "as of" timestamp. No silent staleness.

8.1 Kill switches

Global funding	Halts all new advances instantly. Two-person authorisation, effective in under 10 seconds, with full audit of who, when and why.
Per engine	Dynamic discounting, escrow and treasury are independently switchable. One engine's problem never takes down the others.
Per tenant	Suspends a single SME pending fraud or compliance review, with a mandatory stated reason recorded.
Per buyer	Stops financing a specific buyer's invoices on deterioration, without affecting any other relationship those suppliers hold.
Per sector	Rule-pack-level tightening or suspension by sector code, promotable in minutes.
Per rail	Diverts away from a degraded payment rail to a declared alternative.
Automatic guardrails	Loss rate, dispute rate, duplicate-payout detection, reconciliation break age and error-budget burn each trigger their declared switch without waiting for a human.

Every switch is drilled. Monthly in staging, quarterly in production. A switch that has not been exercised in 90 days is treated as broken and flagged on the readiness dashboard.

8.2 Degrade-to-manual

When automation cannot proceed, work moves to a human queue inside the product rather than stopping. The operator console shows the decision record, the blocking rule, the full feature snapshot and the available actions, and every manual action is written to the same append-only audit store with the operator's identity attached. Manual handling is a declared mode of the system, not an absence of the system.

PART 9

Backtesting and model governance

9.1 Shadow pricing

Before any rule pack prices a real dirham it is evaluated against twelve months of du's historical invoice and payment data, producing for every historical invoice the limit and price the candidate pack would have set and the outcome that actually occurred.

Outputs	Approval rate by segment, price distribution, expected versus realised loss, concentration profile, and the count of decisions that would have changed versus the active pack.
Acceptance	A candidate must not increase modelled loss beyond the declared band, must not reduce approvals in any protected segment without a documented rationale, and must explain every outcome change by a named rule.
Live shadow	After historical shadow passes, the candidate runs in parallel on live traffic for 30 days, deciding nothing and recording everything.

9.2 Forecast backtesting

Method	Walk-forward validation with an expanding window. No random splits — cash flow is a time series and random splits leak the future, which is the most common way a forecast looks good and performs badly.
Horizons	7, 14, 30 and 90 days, each scored separately because they are used for different decisions.
Metrics	MAPE and weighted MAPE on inflow; pinball loss at the 10th, 50th and 90th percentiles, since the product uses the lower tail for the sweep decision; obligation-detection recall, where a miss is far costlier than a false positive.
Segmented reporting	By sector, size band and tenure. An aggregate number conceals exactly the segments where the model is unfit, so aggregate-only reporting is not accepted.
Champion-challenger	A challenger must beat the champion on the primary metric across at least four of five sectors and not regress materially on any, over two consecutive evaluation windows.

9.3 Drift and monitoring

- **Feature drift:** population stability index on every input feature, weekly. PSI above 0.25 opens an investigation automatically.
- **Calibration drift:** realised against predicted, weekly, by segment. Sustained degradation widens the treasury safety margin automatically before anyone intervenes.
- **Outcome drift:** realised loss against modelled band, continuously. A breach triggers mode M1 without human delay.
- **Fairness dispersion:** approval and price distribution by sector, size and emirate each wave, with unexplained dispersion raised to the risk committee as a standing agenda item.
- **Replay regression:** nightly job replays a stratified sample of historical decisions against their recorded pack version and asserts byte-identical outcomes. Any divergence fails the build and blocks promotion.

9.4 Model inventory

MODEL	USE	DECISION AUTHORITY	REVIEW
Cash-flow forecast	Projection, sweepable amount, safety margin	Advisory. Feeds rules; decides nothing.	Monthly
Obligation detection	Payroll, rent, VAT and recurring outflow calendar	Advisory, tuned for recall over precision	Monthly
Buyer payment behaviour	Buyer tier input to limits and pricing	Advisory. The rule pack sets the cap.	Quarterly
Fraud anomaly score	Routing to review, never auto-rejection	Advisory. A human decides every fraud outcome.	Monthly
Collection risk rank	Operational prioritisation only	No customer-facing effect whatsoever	Quarterly

No model in the inventory has decision authority. That is the governance position, and it is enforced architecturally: `ruleengine` accepts model outputs only as named features inside a hashed snapshot, and cannot call a model.

PART 10

Security, privacy and compliance

DOMAIN	CONTROL
Data residency	All personal and financial data stored and processed in UAE regions. No cross-border transfer of SME financial data. Backups and disaster recovery remain in-country.
Encryption	TLS 1.3 in transit; AES-256 at rest with envelope encryption. Keys in a managed HSM-backed KMS, rotated annually, with separate keys per environment and no production key ever present in a lower environment.
Field-level protection	Bank account numbers, UBO identifiers and tax IDs encrypted at field level with a separate key and a separate access grant. Decryption is logged as a discrete audited event.
Access control	Least privilege, time-bound elevation with mandatory justification, two-person rule for kill switches, rule-pack promotion and any production data access. No standing production access for engineers.
Audit	Append-only store, write-once retention, legal hold, tamper-evident chaining. Covers every decision, every money movement, every consent change, every operator action and every data access.
PDPL	Purpose limitation enforced by the consent service at call time rather than by policy. Data-subject rights — access, correction, erasure where lawful, portability — served through self-service in-product.
The commitment	SME financial data is never sold and never used for marketing. Enforced by access control and audited, not merely stated in a policy document.
AML and sanctions	KYB with trade licence and establishment card, UBO screening, sanctions and PEP screening on SME and buyer, ongoing transaction monitoring, SAR filing through du's existing compliance function.
Secure development	SAST and dependency scanning on every commit, secret scanning with push protection, signed commits, SBOM per release, annual third-party penetration test with findings tracked to closure before the next release train.
Incident response	Severity matrix, 15-minute page for sev-1, documented communication tree including regulatory notification timelines, blameless post-incident review published internally within five working days.

PART 11

Design system

WATERLINE is a financial instrument that lives inside someone else's app. It must feel native to du Business, trustworthy enough to move money through, and calm enough to use at 11pm when payroll is due on Thursday.

11.1 Design principles

- **Money is never decorative.** Amounts are the largest, clearest element on any screen they appear on. Nothing competes with them.
- **Show the arithmetic.** Fees are shown as days early × rate × principal, in dirhams, before and after acceptance. Never a single opaque number.
- **State before action.** Every surface tells the SME where its money is before offering to move it.
- **Calm by default, loud only when it matters.** The accent colour is reserved for action and positive value. Red appears only for genuine failure, never for ordinary validation.
- **Honest latency.** Processing states show what is happening and what remains, with real progress where it exists and an honest indeterminate state where it does not.
- **Arabic is a first-class language, not a translation layer.** RTL is designed, not mirrored as an afterthought.

11.2 Colour



Semantic mapping. `accent` is action and positive liquidity. `success` confirms settled money only — never a saved preference. `caution` marks attention needed, used for expiring licences and approaching obligations. `danger` is reserved for genuine failure and irreversible actions. `escrow` and `treasury` are engine identity colours used sparingly, for wayfinding only, never for emphasis.

Dark mode is a full re-pairing, not an inversion: `paper` becomes `#0B1420`, surfaces step up through `#141C28` and `#1C2634`, and `accent.500` lightens to `#4D9BE8` to hold contrast on dark. Every pair is verified at AA before release.

11.3 Gradients

liquidity.fill	`linear-gradient(180deg, #2783DE 0%, #1B5FA3 100%)` — the only gradient permitted on a primary action. Used on the fund button and the liquidity gauge fill.

surface.lift	`linear-gradient(180deg, #FFFFFF 0%, #FAF8F5 100%)` — a 2% separation for elevated cards, replacing a heavy shadow.
waterline.horizon	`linear-gradient(180deg, rgba(39,131,222,0.14) 0%, rgba(39,131,222,0) 72%)` — the signature motif behind the balance header, suggesting a level without drawing a chart.
skeleton.shimmer	`linear-gradient(100deg, #E8EBEF 18%, #F3F5F8 32%, #E8EBEF 46%)` animated 1.4 s, a single pass, left to right and mirrored in RTL.
escrow.lock	`linear-gradient(135deg, #6B4FA8 0%, #4A3579 100%)` — escrow contract header only.

Gradients never appear on text, never on more than one element in a viewport, and never exceed a 2-stop definition. A third stop is a design review failure.

11.4 Typography

Latin: Inter, with `tnum` and `lnum` enabled globally so digits align in every column. **Arabic:** IBM Plex Sans Arabic, selected for its structural match to Inter at shared optical sizes. **Monospace:** JetBrains Mono, used only for references, hashes and identifiers. System fallbacks are declared and tested, never assumed.

`display / 34px / 700 / -2.2%`	**AED 142,800**
`amount.lg / 28px / 700 / -1.8%`	**AED 42,000**
`title / 20px / 700 / -1.2%`	**Get paid today**
`heading / 17px / 600 / -0.6%`	**Three invoices ready to fund**
`body / 15px / 400 / 0 / 1.55`	Your buyer confirmed this invoice on 12 October.
`body.strong / 15px / 600`	**Net proceeds AED 41,286**
`caption / 13px / 400`	Settles to du Pay ending 4471
`label / 11px / 700 / +14% / upper`	READY TO FUND
`mono / 13px / 400`	`dec_01J9Z4K2QX`

Scale is a 1.2 modular ratio snapped to 1px. Line length is capped at 68 characters for body copy. Amounts never wrap and never truncate — the container adapts, or the type steps down one level, but a figure is never shown partially.

11.5 Grid, spacing, radius, elevation

Spacing	4px base. Scale 4 / 8 / 12 / 16 / 24 / 32 / 48 / 64. No arbitrary values; a non-scale value fails lint.
Grid	Mobile 4-column, 16px gutter, 20px margin. Tablet 8-column. Desktop 12-column, 24px gutter, 1240px max content width.
Radius	sm 6px (inputs, chips) · md 10px (cards) · lg 16px (sheets, modals) · full (avatars, pills). Nothing is square except table cells and dividers.

Elevation	0 flat · 1 `0 1px 2px rgba(11,20,32,.06)` cards · 2 `0 4px 12px rgba(11,20,32,.08)` popovers · 3 `0 12px 32px rgba(11,20,32,.14)` modals. Four levels only; a fifth is a sign the hierarchy is wrong.
Borders	1px `ink.100` default, `ink.200` on hover, 2px `accent.500` on focus. Focus is never removed, only restyled.

11.6 Icons and glyphs

Style	Custom 24px grid, 1.75px stroke, round caps, round joins, 2px optical inset. Outline by default; filled variant only for the active navigation state.
Core set	invoice, invoice-verified, invoice-disputed, fund, escrow-lock, milestone, treasury-sweep, treasury-return, buyer, consent, audit-record, rule-pack, status-degraded, kill-switch.
Signature glyphs	**The waterline** — a horizontal rule with a 2px fill below, marking a level. Used for the balance header, the liquidity gauge and the empty state, and nowhere else. **The gauge** — a vertical measuring column showing available against total, the product's single most recognisable element.
Currency	AED is set in type, never as an icon, so it inherits tabular figures and aligns in columns.
Status dots	8px, paired always with a text label. Colour alone never conveys state, for accessibility and for printing.
Delivery	Single SVG sprite, `currentColor` throughout, 1-year immutable cache keyed by content hash. No icon font.

11.7 Navigation

Entry	WATERLINE is a tile inside du Business. It never competes with du's own navigation and never attempts to replace it.
Mobile	Four-tab bottom bar: Cash, Invoices, Escrow, Activity. Labels always visible — icon-only navigation fails users under stress, which is exactly when this product is used.
Desktop	Left rail, 240px, collapsible to 64px icons with persistent tooltips. Section state persists per user.
Hierarchy	Maximum two levels below a tab. A third level is a panel or a sheet, never a new navigation tier.
Back and dismiss	Every modal and sheet is dismissible by Escape, by backdrop tap and by an explicit control. Browser back closes the topmost layer rather than leaving the product.
Deep links	Every invoice, offer, contract and decision has a stable URL that survives authentication, so support can send an SME directly to the thing being discussed.
Breadcrumbs	Desktop only, from the second level down, with the current item unlinked.

11.8 Widgets

Liquidity gauge	The signature widget. Vertical column showing available liquidity against total approved receivables, with the waterline marking the floor. Animates on value change over 420 ms with easing, never on mount.
Cash runway strip	Horizontal 90-day forward strip with obligation markers for payroll, rent and VAT. Tap a marker to see the detected obligation and its confidence.
Invoice card	Buyer, amount, due date, verification state, and a single primary action. Swipe on mobile for secondary actions; never more than one primary per card.
Fee breakdown	Always expanded, never behind a disclosure. Days early, rate per day, fee in dirhams, net proceeds, each on its own line.

Date slider	Settlement-date selector recomputing fee locally against the cached rate curve. Haptic detent at each day on mobile.
Milestone tracker	Vertical stepper with state per milestone, evidence links, approval actor and timestamp.
Decision receipt	Plain-language reason, rule-pack version, decision reference, and a copy control. Attached to every outcome including rejections.
Status banner	Rendered directly from the degrade-mode endpoint. The UI never guesses system state.

11.9 Modals, panels and sheets

Modal	Blocking, for irreversible confirmation only: accept an offer, approve a milestone, revoke consent. Maximum 480px, focus trapped, Escape closes, primary action right in LTR and left in RTL.
Sheet	Mobile bottom sheet with two detents, 50% and 92%. Drag to dismiss with velocity threshold. Used for fee detail, buyer detail and filters.
Side panel	Desktop right panel, 420px, non-blocking so the list stays visible and navigable. Used for invoice detail and decision records.
Popover	Inline explanation anchored to a control, 280px max, dismissed on outside interaction. Used for "why this limit" and freshness indicators.
Toast	Non-blocking confirmation, 4 s, bottom-centre mobile and bottom-right desktop. Never used for errors that need action, and never for money movement, which gets a persistent record instead.
Layer discipline	One blocking layer at a time. A modal never opens from a modal; the second becomes a step inside the first.

11.10 Loading states

< 200 ms	Nothing. A flash of a spinner is worse than a brief wait.
200 ms – 1 s	Skeleton matched to final layout within 4px, so content does not shift on arrival. Shimmer is a single pass, 1.4 s, mirrored in RTL.
1 s – 10 s	Skeleton plus a named stage: "Checking with your buyer", "Pricing your invoice". Never a bare "Loading".
> 10 s	Stepped progress with completed stages ticked and the current stage active, plus an honest expected time and the option to be notified rather than wait.
Optimistic	Permitted for preference changes only. Never for money. A payout never shows success before the ledger confirms it.
Stale data	Last-known-good is shown with a visible "as of" timestamp and a refresh control, rather than a blank screen.

11.11 Error states and boundaries

Inline field	Below the field, `danger.500`, with the correction stated: "Enter the amount in AED, without decimals." Never "Invalid input."
Form level	Summary at the top listing each failed field as a focusable link, announced to assistive technology.
Rule rejection	Not styled as an error. It is a neutral outcome card with the plain-language reason and the specific path to becoming eligible.
Transaction failure	Full-width card with what happened, whether money moved, what happens next, and the trace reference. "Your money did not leave your account" is always stated explicitly when true.

Degraded service	Persistent banner derived from the status endpoint, naming the affected capability and what still works.
Component boundary	Every widget is wrapped. A failed widget renders a compact inline fallback with a retry control; the rest of the screen continues to work. One broken chart never blanks a dashboard.
Route boundary	Preserves navigation and offers return to Cash. The user is never stranded on a blank screen with no way out.
Root boundary	Branded failure page with a trace reference and a direct support path. Logged with full context before render.
Offline	Cached read-only view with an explicit offline indicator. Write actions are disabled with the reason given, never silently failing.

11.12 Motion

```
durations   instant 90ms · quick 160ms · base 240ms · deliberate 420ms · slow 640ms
easing      standard  cubic-bezier(.20,0,0,1)     enter/exit
            decelerate cubic-bezier(0,0,0,1)      incoming
            accelerate cubic-bezier(.30,0,1,1)    outgoing
            spring     stiffness 210, damping 26  gauge, sheets, drag
reduced     prefers-reduced-motion: all transforms → opacity-only crossfade 90ms,
            shimmer static, parallax disabled, gauge value snaps
```

MICRO-INTERACTIONS

- Button press: scale 0.98, 90 ms, with haptic light impact on mobile.
- Toggle: 160 ms slide with a 1.06 overshoot on the knob only.
- Slider detent: 90 ms colour shift plus haptic selection tick per day.
- Field focus: 160 ms border colour and 2px ring, never a layout shift.
- Copy reference: glyph morphs to a tick for 1.2 s, then reverts.
- Amount change: digit roll on the changed digits only, 240 ms, tabular figures so nothing reflows.

MACRO-CHOREOGRAPHY

- **Funding confirmation:** amount lifts 8px and scales to 1.04 (240 ms) → gauge fill rises to the new level (420 ms, spring) → receipt card enters from below (240 ms, decelerate) → success glyph draws its stroke over 320 ms. Total under 1.1 s, and the sequence is skippable by tapping.
- **Sheet presentation:** backdrop fades to 40% over 160 ms while the sheet rises with spring; content fades in 60 ms after the sheet starts, so the container leads and the content follows.
- **Route transition:** 160 ms crossfade with a 12px directional offset. Forward moves with reading direction, back against it, mirrored under RTL.
- **List reorder:** FLIP technique, 240 ms, staggered 20 ms per row to a maximum of 8 rows, then simultaneous.

SCROLL BEHAVIOUR

- Balance header collapses from 180px to a 56px sticky bar across the first 120px of scroll, with the amount scaling and the waterline gradient fading. Interpolated to scroll position, not triggered at a threshold, so it tracks the finger exactly.
- Section labels stick with a 1px rule that appears only once content passes beneath.
- Reveal on scroll is limited to 12px translate and opacity, triggered at 15% visibility, once per element per session. No long parallax, no pinned sections, no scroll-jacking — this is a financial tool, not a landing page.
- Lists are virtualised above 60 rows with scroll-position restoration on back navigation.

- Pull to refresh on mobile with a resistance curve and a haptic tick at the trigger point.

11.13 Processing and money-in-motion

The most trust-sensitive moment in the product is the gap between accepting an offer and seeing the money. It is designed explicitly rather than left to a spinner.

Stage display	Four named stages — Accepted, Reserved, Sent, Settled — each with a real timestamp as it completes. The SME always knows exactly where in the pipeline their money is.
Honest indeterminate	When the provider gives no progress signal, the stage shows elapsed time and the expected window rather than a fake progress bar. Fabricated progress is a trust failure.
Ambiguous outcome	Stage shows "Confirming with the bank" with a statement that no duplicate payment is possible, because the idempotency guarantee is the user's reassurance, not just an engineering property.
Background completion	The SME can leave. Push notification on terminal state, and the activity record is authoritative regardless of whether the app was open.
Never a dead end	Every processing state shows elapsed time, expected completion, and a support path with the trace reference pre-filled.

11.14 Accessibility and localisation

- WCAG 2.2 AA throughout; AAA on amounts and all critical actions. Verified in CI with axe, and manually with VoiceOver and TalkBack each release.
- Minimum touch target 44×44px with 8px separation. Primary actions sit inside thumb reach on mobile.
- Full keyboard operation with a visible focus ring, logical tab order, and a skip link to main content.
- Screen-reader announcement of every state change: processing stage, outcome, error. Amounts announced as "forty-two thousand dirhams", not digit by digit.
- Arabic and RTL as a designed mirror: layout, icons with direction, progress and slider direction, with Eastern Arabic numerals as a user preference and Western as default.
- Text scales to 200% without loss of function. Amounts step type level rather than truncate.
- Colour never carries state alone — every status dot has a label, every chart series a pattern.

PART 12

Observability, testing and release

12.1 Observability

SLOs	Funding path availability 99.9%. Quote p99 400 ms. Payout initiation p99 1.2 s. Acceptance-to-settled median under 5 minutes, p95 under 1 hour. Error budget burn reviewed weekly; a burn above 2× pauses feature releases automatically.
Golden signals	Per service: rate, errors, duration, saturation. Per business flow: quotes issued, offers accepted, payouts settled, straight-through rate, exception queue depth.
Tracing	100% sampling on the funding path, 5% elsewhere. Every trace carries tenant, decision ID and rule-pack version as attributes, so any transaction is reconstructable end to end.
Structured logs	JSON, correlated by trace ID, with PII redacted at the logging library rather than by convention. A log line containing an unredacted account number fails the build.
Alerting	Symptom-based, never cause-based. Pages only on user-visible impact or integrity risk. Every alert has a runbook link; an alert without a runbook is deleted.
Business dashboards	Funded volume, loss rate, dispute rate, DSO delta, reconciliation breaks, forecast calibration, fairness dispersion — reviewed weekly with the du pilot team.

12.2 Testing strategy

Unit	Rule pack evaluated against a golden fixture set per version. Every rule has a passing and a failing case plus boundary cases at each threshold.
Property-based	Ledger invariants — transactions always balance, balances never derive negative, idempotency always holds — asserted against generated inputs, not hand-picked examples.
Replay regression	Nightly replay of a stratified historical decision sample, asserting byte-identical outcomes against the recorded pack version. Any divergence blocks promotion.
Contract	Consumer-driven contracts for every external dependency, verified against recorded provider responses including every documented error shape.
Integration	Full funding path against provider sandboxes, including every failure injection: timeout, 500, duplicate callback, out-of-order callback, lost callback.
End-to-end	Playwright across the critical journeys on mobile and desktop, LTR and RTL, light and dark, plus the axe accessibility suite.
Chaos	Scheduled fault injection in staging: kill a zone, saturate a pool, delay a provider, corrupt a callback. Each exercise must demonstrate the declared degrade mode engaging correctly.
Load	3× projected peak for the full 40,000-account base, sustained for one hour, with p99 budgets held and no queue growth.
Operational drills	Kill switches monthly, zone failover quarterly, reconciliation-break response quarterly, incident comms annually with du's communications team.

12.3 Environments and release

- Four environments: local, integration (provider sandboxes), staging (production-shaped, synthetic data only), production. No production data ever enters a lower environment, including for debugging.
- Trunk-based development with short-lived branches. Every change behind a flag. Flags are inventoried with an owner and an expiry date; an expired flag fails the build.
- Progressive delivery: 1% → 5% → 25% → 100% with automatic rollback on SLO or guardrail breach at each step.

- Database migrations are expand-contract only, forward-compatible across one release, and never destructive in the same release as the code that stops using the column.
- Rule-pack releases are independent of code releases and roll back by pointer change in under 60 seconds.
- Every release is reproducible from a signed commit with an SBOM attached.

PART 13

Production readiness checklist

No item is optional. Nothing ships to a funded wave with an unticked box, and the list is reviewed as a gate rather than as a reminder.

CORRECTNESS AND MONEY

- [ ] **Ledger balance** Every transaction sums to zero, enforced by database constraint and asserted by property tests.
- [ ] **No floats** Integer minor units throughout; a lint rule fails any float in a monetary path.
- [ ] **Idempotency** Required and enforced on every money-moving endpoint; duplicate payout proven impossible under test.
- [ ] **Reconciliation** Daily to the dirham against provider statements, with automated break detection and same-day escalation.
- [ ] **Rounding** Residuals posted to a declared account and reconciled, never absorbed.
- [ ] **Replay** Historical decisions replay byte-identically against their recorded pack version.

RESILIENCE

- [ ] **Timeouts** Explicit on every network call, query and lock; missing timeout fails the build.
- [ ] **Backoff** Full jitter on every retry class, with retry budgets enforced at the client.
- [ ] **Breakers and bulkheads** Configured per dependency with isolated pools; state exposed on the status endpoint.
- [ ] **Outbox** Every state change publishes its event transactionally; orphan detection runs continuously.
- [ ] **DLQ** Every topic has one, alerting at 15 minutes non-empty, with tested bulk replay.
- [ ] **Degrade modes** M0–M4 implemented, drilled, and driving the UI from the status endpoint.
- [ ] **Kill switches** All six scopes implemented, two-person authorised, exercised within the last 30 days.
- [ ] **Failover** Zone failover drilled within RTO, with results published internally.
- [ ] **Load** 3× full-base peak sustained one hour with budgets held.

SECURITY AND COMPLIANCE

- [ ] **Residency** All data in UAE regions including backups; verified, not assumed.
- [ ] **Encryption** TLS 1.3, AES-256 at rest, field-level on sensitive identifiers, keys rotated.
- [ ] **Access** No standing production access; elevation time-bound and justified; two-person rule on sensitive operations.
- [ ] **Audit** Append-only, tamper-evident, covering decisions, movements, consent, operator actions and data access.
- [ ] **Consent** Enforced at call time; revocation propagates within 60 s and purges derived data within 24 h.
- [ ] **Screening** KYB, UBO, sanctions and PEP at onboarding and on change, with ongoing monitoring.
- [ ] **Pen test** Third-party test complete with all high and critical findings closed.
- [ ] **Legal** Assignment enforceability and product characterisation opinions obtained before Wave 1.

EXPERIENCE

- [ ] **Accessibility** WCAG 2.2 AA verified in CI and manually with screen readers each release.
- [ ] **RTL** Arabic reviewed by a native speaker in product context, not in a spreadsheet.
- [ ] **Reduced motion** Honoured across every animation including the gauge and shimmer.
- [ ] **Boundaries** Component, route and root boundaries implemented with tested fallbacks.
- [ ] **Offline** Read-only cached view with explicit indicator; writes disabled with a stated reason.
- [ ] **Copy review** Every error and rejection reviewed for plain language and an actionable next step.

OPERATIONS

- [ ] **Runbooks** One per alert; every alert without a runbook is deleted rather than tolerated.
- [ ] **On-call** Rota staffed with escalation path, 15-minute sev-1 page, and a tested paging channel.
- [ ] **Exception queue** Staffed with response-time commitment; SME notified proactively on every exception.
- [ ] **Support tooling** Agents can locate any transaction from a trace reference and see the full decision record.
- [ ] **Status page** Public degrade-mode status with incident history.
- [ ] **Flag hygiene** Every flag owned with an expiry; expired flags fail the build.

PART 14

Must-haves

The three non-negotiables from the brief, mapped to the specific mechanism in this document that satisfies each.

REQUIREMENT	HOW IT IS MET	WHERE
Built for rapid scalability	Stateless funding path, per-tenant hash partitioning with row-level security, separated hot/warm/cold paths, active-active across two UAE zones, 85% straight-through so operations scale sub-linearly. Proven by a 3× full-base load test and a drilled failover before the pilot concludes.	Parts 2.2–2.4, 6, 12.2
100% digital integration capability	Every lifecycle step digital end to end — UAE Pass identity, consent, verification, pricing, acceptance, payout, collection, reconciliation, dispute, reporting, offboarding. Exceptions are handled by humans inside the same system with the same audit trail. No paper, no branch, no spreadsheet of truth.	Parts 4, 8.2, 11
Scalable to 40,000+ SME customers	Pilot and full base run the same binary, the same rule pack and the same audit store, differing only by feature flag and tenant count. There is no pilot build to rewrite. Rails have headroom: Aani settles 24/7 and du Pay has transacted AED 1.5bn+.	Parts 2, 3.3, 12.3

PART 15

Master build prompt

Copy this prompt verbatim into a coding agent. It is written to be executed end to end without omission, and it refers back to this document for every specification it does not restate.

> **HOW TO USE IT**
>
> Supply this PRD alongside the prompt. The prompt instructs the agent to treat the PRD as binding and to stop rather than invent when a specification is genuinely absent. Execute phases in order; do not allow the agent to skip ahead to a later phase because an earlier one is "mostly done".

```
ROLE
You are a staff-level engineer building WATERLINE, a production financial
system that moves real money for small businesses in the UAE. You are
building it for du (EITC) under the Ignyte SME Challenge, Theme 2 —
Creating Liquidity. Treat every instruction below as binding.

MISSION
Build the WATERLINE liquidity layer to production standard: dynamic
discounting on buyer-approved receivables, milestone escrow via
programmable mandate at a licensed custodian, and treasury autopilot
sweeping idle cash into low-risk short-term instruments. It embeds in the
du Business app, settles over du Pay and Aani, and must scale to 40,000+
SME tenants without a rewrite.

NON-NEGOTIABLES — violating any of these is a build failure
1.  No model decides who gets paid. A deterministic, versioned rule pack
    decides eligibility, limits and prices. ML produces named features
    inside a hashed snapshot and nothing else. ruleengine performs no I/O,
    reads no clock and uses no randomness.
2.  Every decision writes an append-only record BEFORE the outcome is
    shown: input hash, rulepack version, resolved parameters, outcome,
    binding rule, reason code, plain-language reason. If the write fails,
    the outcome is not shown.
3.  All money is integer minor units. No floating point touches a
    monetary path anywhere. Enforce with a lint rule that fails the build.
4.  Idempotency-Key is required on every money-moving endpoint. Duplicate
    payout must be structurally impossible, including across process
    restarts and ambiguous provider timeouts.
5.  Read-only on du BSS. No write path in any phase. Never in the rating
    or charging path. Total WATERLINE unavailability must have zero
    effect on du billing.
6.  The SME's treasury floor is absolute. No optimisation, yield
    opportunity or operator override may breach it. Enforce as a hard
    stop in ruleengine and make it the most heavily tested invariant in
    the system.
7.  Never fail silently on money. Ambiguity on the funding path resolves
    to not paying, then reconciling. Ambiguity on a read path resolves to
    last-known-good with a visible freshness indicator.
8.  Pilot and full base run the same binary, the same rule pack and the
    same audit store. Differences are feature flags and tenant count
    only. Do not build a pilot-only path.

STACK
TypeScript throughout. Node 22 services (Fastify), React 19 + Vite for
the embedded SDK, React Native for mobile surfaces. Postgres 16 as system
of record with row-level security. Redis for cache and single-flight.
Kafka or equivalent for the event log. OpenTelemetry for traces, metrics
and logs. Terraform for infrastructure, two UAE availability zones,
active-active. If you must deviate, state the reason and the trade-off
before writing code.
```

```
BUILD ORDER — execute in sequence, do not skip, do not reorder

PHASE 0 — Foundations
  * Monorepo, workspaces, strict tsconfig, lint rules including the
    no-float-in-money rule and the mandatory-timeout rule.
  * CI: typecheck, lint, unit, property, contract, e2e, axe, SBOM.
  * Postgres schema exactly as specified in PRD Part 2.5, including the
    balanced-transaction constraint, the global invoice_hash index, the
    insert-only grant on decision, and row-level security on every table.
  * Terraform for both zones, with synchronous replication for ledger and
    decision stores and asynchronous for everything else.
  ACCEPTANCE: a migration run produces an empty but complete schema; a
  query without tenant context returns zero rows; CI is green.

PHASE 1 — Rule engine and decision service
  * ruleengine as a pure function: (featureSnapshot, rulePack) => Decision.
    No I/O, no clock, no randomness. Reject at the type level.
  * Rule pack loader with signature verification, version pinning and
    pointer-based activation. Prior versions remain loadable forever.
  * Canonical feature-snapshot serialisation and hashing.
  * decision service: assemble features, evaluate, persist record, emit
    event via transactional outbox.
  * Golden fixture suite: every rule with a passing case, a failing case
    and boundary cases at each threshold.
  * Replay endpoint and nightly replay regression job.
  ACCEPTANCE: replaying any stored input_hash against its recorded
  rulepack_version produces a byte-identical outcome, asserted in CI.

PHASE 2 — Ledger and payout
  * Double-entry ledger. Balances derived, never stored as truth.
    Property tests for: transactions always balance, derived balances
    never negative, posting is idempotent on (tenant_id, source_event_id).
  * Transactional outbox and relay. Orphan detection for both
    directions: state change without event, event without state change.
  * payout service over du Pay and Aani with provider-facing idempotency
    keys derived deterministically from offer_id.
  * Retry policy exactly as PRD Part 6.1, including full jitter, retry
    budgets and deadline propagation. One central implementation; per-call
    overrides are forbidden.
  * Circuit breakers and per-dependency bulkheads as PRD Part 6.6.
  * Daily reconciliation against provider statements with automated break
    detection and same-day escalation. Rounding residuals to a declared
    account.
  ACCEPTANCE: inject timeout, 500, duplicate callback, out-of-order
  callback and lost callback; in every case the ledger is correct, no
  duplicate payment occurs, and reconciliation closes to zero.

PHASE 3 — Invoice, consent, registry
  * Invoice state machine exactly as PRD Part 4.2 including canonical
    hashing, cross-tenant duplicate detection, and verification priority
    with per-route advance-rate caps.
  * Consent service with per-purpose scoped grants, call-time enforcement,
    60-second revocation propagation and 24-hour derived-data purge.
  * Registry: KYB, trade licence with expiry handling, UBO, buyer links
    with tier derivation.
  * Buyer confirmation portal: single-click confirm, bulk confirm, and an
    API callback route for anchor ERP systems.
  ACCEPTANCE: a deliberate duplicate submitted by a second tenant freezes
  both submissions and routes to review without auto-rejecting either;
  revoking consent disables dependent features within 60 seconds.

PHASE 4 — Engines
  * Dynamic discounting: quote, offer, acceptance with server-side
    re-validation, reservation, funding, assignment, collection.
    A price change between quote and acceptance is surfaced explicitly
```

```
    and requires fresh acceptance. Never apply a changed price silently.
  * Milestone escrow: contract creation, the three condition types,
    custodian release instruction, dispute freeze on the disputed
    milestone only, termination with full history export.
  * Treasury autopilot: obligation calendar, sweepable amount
    (balance − floor − obligations(horizon) − safety_margin), instrument
    eligibility from rule-pack data, automatic recall with margin, and
    the absolute floor guarantee.
  ACCEPTANCE: every edge case in PRD Part 7 has a test that demonstrates
  the specified behaviour. Not a similar behaviour. The specified one.

PHASE 5 — Degrade modes and kill switches
  * Modes M0–M4 exactly as PRD Part 8, with automatic transition on
    guardrail breach and manual operator transition.
  * All six kill-switch scopes with two-person authorisation and full
    audit, effective in under 10 seconds.
  * Status endpoint publishing current mode and breaker state; the UI
    reads system state from it and never guesses.
  * Degrade-to-manual: operator console showing the decision record, the
    blocking rule, the full feature snapshot and the available actions,
    with every manual action written to the same audit store.
  ACCEPTANCE: chaos exercises — kill a zone, saturate a pool, delay a
  provider, corrupt a callback — each demonstrably engage the correct
  declared mode, and the UI reflects it without code change.

PHASE 6 — Experience
  * Implement the design system in PRD Part 11 completely: colour tokens
    including dark-mode re-pairing, the five permitted gradients, the
    type scale with tabular figures, the 4px spacing scale, the four
    elevation levels, the custom 24px icon grid, and both signature
    glyphs (the waterline rule and the vertical gauge).
  * Navigation, widgets, modals, sheets, side panels, popovers and toasts
    exactly as specified, including the one-blocking-layer rule.
  * Loading states by duration band: nothing under 200 ms, layout-matched
    skeletons to 1 s, named stages to 10 s, stepped progress beyond.
    Optimistic updates for preferences only, never for money.
  * Error states and all three boundary levels — component, route, root —
    with tested fallbacks. A rule rejection is a neutral outcome card,
    not an error.
  * Motion: the full duration and easing table, every micro-interaction,
    all four macro-choreographies, the scroll-linked balance header, and
    complete prefers-reduced-motion handling.
  * Processing states: four named stages with real timestamps, honest
    indeterminate display, explicit no-duplicate-payment reassurance on
    ambiguous outcomes, and background completion with push.
  * Accessibility: WCAG 2.2 AA with AAA on amounts and critical actions,
    full keyboard operation, screen-reader announcement of every state
    change, 200% text scaling.
  * Arabic and RTL as a designed mirror, with Eastern Arabic numerals as
    a user preference.
  ACCEPTANCE: axe passes with zero violations; VoiceOver and TalkBack
  complete the full funding journey; RTL reviewed in product context;
  reduced-motion verified on every animation including gauge and shimmer.

PHASE 7 — Scale and hardening
  * Hot/warm/cold path separation verified: no hot-path request makes a
    synchronous external call, ever.
  * Caching exactly as PRD Part 6.4, including the never-cached list,
    single-flight, probabilistic early expiry and tenant-scoped keys.
  * Queues as PRD Part 6.3: declared partition keys, honest ordering
    guarantees, DLQs with 15-minute alerting and tested bulk replay,
    poison-message routing after three failures, declared backpressure
    priority.
  * Active-active completion with home-zone assignment per tenant so two
    zones never write the same ledger account concurrently.
  * Load test at 3x projected peak for 40,000 accounts, sustained one
```

```
    hour, p99 budgets held, no queue growth.
  * Zone failover drill inside RTO.
  ACCEPTANCE: PRD Part 13 production readiness checklist fully ticked,
  with evidence linked for every item.

CODING STANDARDS
* Typed errors only. One top-level catch-all per service, at the request
  boundary, which logs with trace ID, increments a counter and returns
  internal_error. Every other catch handles a declared type.
* Never swallow an error. Handle with a declared recovery or rethrow with
  added context. Log-and-continue is forbidden on any money path.
* Explicit timeout on every network call, query and lock acquisition.
* Compensating transactions written and tested as first-class code paths,
  never as cleanup scripts.
* Every error carries tenant, subject reference, decision ID where
  applicable, and trace ID.
* PII redacted in the logging library, not by convention.
* Expand-contract migrations only, forward-compatible across one release.
* Every change behind a feature flag; every flag owned with an expiry.

DEFINITION OF DONE — per phase, all of these
  [ ] Specified behaviour implemented, including every edge case named
      in the PRD for that area
  [ ] Unit, property and contract tests passing
  [ ] Failure injection demonstrating the declared degrade behaviour
  [ ] Observability: traces, metrics, structured logs, and an alert with
      a runbook for every user-visible failure mode
  [ ] Audit records written for every decision and money movement
  [ ] Accessibility verified where UI is involved
  [ ] Documentation updated: API reference, runbook, architecture note

PROHIBITIONS
* Do not add an unsecured lending product, consumer credit, FX, crypto
  custody or insurance. All are explicitly out of scope.
* Do not let a model make or override a credit decision under any
  framing, including "advisory override" and "human in the loop with a
  default".
* Do not write to du BSS. Do not enter the rating or charging path.
* Do not cache balances, limits at acceptance, consent state, KYB state
  or kill-switch state.
* Do not implement tokenised revenue in Phases 1 and 2. It is Phase 3,
  feature-flagged, pending CBUAE and VARA guidance.
* Do not substitute a default for a missing feature. Reject with a typed
  reason code.
* Do not show success for a money movement before the ledger confirms it.
* Do not fabricate progress. An honest indeterminate state is required
  where the provider gives no progress signal.

WHEN A SPECIFICATION IS GENUINELY ABSENT
Stop and ask. Do not invent a credit policy, a fee, a limit, a retention
period, a regulatory interpretation or a threshold. State precisely what
is missing, why it blocks you, and the options you see. Inventing a
number in a financial system is worse than stopping.

OUTPUT FORMAT
For each phase, produce in this order: (1) a short plan naming the files
you will create or change; (2) the complete implementation, no
placeholders, no "TODO", no abbreviated handlers; (3) the tests; (4) the
acceptance evidence showing the phase's acceptance criteria met; (5) a
list of anything you deliberately deferred and why. Then stop and wait
before starting the next phase.
```